

JASMINE STEFFI A

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PROFILE SUMMARY

Biotechnology postgraduate with specialized experimental training in cellular immunology, mammalian cell-based functional assays, and targeted therapeutics. Proven wet-lab track record in quantifying immunomodulatory responses, evaluating expression, and conducting research in international laboratory. Highly motivated to investigate immune and biomaterial interactions to develop next-generation platforms for cancer immunotherapy.

EDUCATION

- M.Tech in Biotechnology** (2026) - Karunya Institute of Technology and Sciences (85.3%, Rank Holder First-Class Distinction)
Thesis: Biosynthesis of nanoparticles, Functional Characterization, and Evaluation of Immunomodulatory and Anti-Inflammatory Responses in Mammalian Macrophage Models.
Focus: Molecular Immunology, Cancer Biology, Advanced Cell Culture Techniques.
- B.Tech in Biotechnology** (2024) - Karunya Institute of Technology and Sciences (84.60%, First-Class Distinction)

INTERNATIONAL RESEARCH EXPERIENCE

Student Exchange Intern, Laboratory of Biomimetic Engineering, University of Chemistry and Technology, Prague, *Jul 2025 - Sep 2025*

- Developed spray-dried microparticles targeting *E. coli* biofilms
- Gained experience in biomaterial-based delivery systems

CORE TECHNICAL COMPETENCIES

Domain	Key Experimental & Analytical Techniques
Cell Culture & invitro Assays	Mammalian cell culture maintenance (RAW 264.7 macrophage lines, cancer models), aseptic cell handling, cryopreservation, cell viability & cytotoxicity assays (MTT, Trypan Blue exclusion), anti-inflammatory/immunomodulatory screening.
Molecular Biology & Biochemistry	Genomic DNA/total RNA extraction, PCR & RT-qPCR primer design/execution, SDS-PAGE, Western blotting, protein extraction and quantification (Bradford/BCA), microencapsulation.
Analytical Platforms	UV-Vis Spectrophotometry, Gas Chromatography-Mass Spectrometry (GC-MS analysis coordination), Fourier-Transform Infrared Spectroscopy (FTIR), Dynamic Light Scattering (DLS).
Informatics & Data Analysis	Bioinformatics sequence alignment (NCBI BLAST), molecular modeling & pathway mapping, GraphPad Prism (statistical testing, dose-response kinetics), BioRender (scientific illustration).

RESEARCH PROJECTS

Biosynthesis of nanoparticles, Functional Characterization, and Evaluation of Immunomodulatory and Anti-Inflammatory Responses in Mammalian Macrophage Models.

- Investigated immunomodulatory and anti-inflammatory properties of biosynthesized therapeutic formulations using *in vitro* mammalian RAW 264.7 macrophage models.
- Conducted quantitative cellular assays to evaluate cell viability, cellular protection, and suppression of inflammatory responses.
- Coordinated specialized analytical characterization (GC-MS metabolite profiling with SITRA) to elucidate chemical constituents driving functional bioactivity.

Non-Canonical Functions of Telomerase in Mitochondria

- Analyzed mitochondrial stress response and mitophagy using mitoKeima assays in U-2 OS osteosarcoma cell line.
- Investigated extra-telomeric roles of Telomerase Reverse Transcriptase (TERT) in mitochondrial function.
- Explored the role of TERT in cancer cell survival and mitochondrial homeostasis.

Alliin-Alliinase Microparticles Against *Escherichia coli* Biofilm

- Developed and evaluated alliin-alliinase microparticle formulations for controlled generation of allicin as an antimicrobial agent.
- Investigated the antibiofilm activity of the microparticles against *Escherichia coli* biofilms using *in vitro* assays.
- Performed microbiological experiments, biofilm quantification, and data analysis to assess treatment efficacy.

Development of Collagen-Based Scaffold for Cell Culture from Marine Waste

- Fabricated collagen-based scaffolds derived from marine waste for *In vitro* cell culture.
- Conducted biomaterial processing and physicochemical characterization.
- Evaluated scaffold suitability for cell adhesion and growth applications.

PUBLICATIONS

- Kanimozhi D, Mathew J, Antu AR, Jasmine Steffi A, Arya K.
An Exploratory In Silico Screening of Potentially Deleterious Single Nucleotide Polymorphisms in the Human CDH1 Gene. International Journal of Drug Delivery Technology. 2026;16(35s):746–759.
DOI: 10.25258/ijddt.16.35s.85

Preprints

- Das S, Dubey A, Laney K, Kompala J, Jasmine Steffi A, Nayar R, et al.
Emodin as a Multi-Target Modulator of Tumor Angiogenesis: Mechanisms, Microenvironment, and Therapeutic Potential. Cambridge Open Engage. 2026. Preprint. DOI: 10.33774/coe-2026-9zbzx
(Submitted to a peer-reviewed journal for publication.)

Manuscripts Submitted

- Jasmine Steffi A, Mathew J, Samuel M.
Xenorhabdus stockiae-Mediated Green Synthesis of Silver Nanoparticles for Anti-Inflammatory Activity.
Manuscript submitted to Journal of Pure and Applied Microbiology.
- Jasmine Steffi A, Jibu Thomas.
Biorefining of Micro- and Macroalgae for Sustainable Biofuels and Resource Valorization.
Final draft submitted to Environmental Science and Pollution Research after reviewers comments

TRAININGS

Azyme Biotech, Bangalore, *Trainee - Microbial Techniques* | June 2023

- Gained hands-on experience in the isolation, culture, and biochemical confirmation of clinically important microorganisms.

Biozone, Chennai, *Trainee - Molecular Biology* | June 2022

- Performed genomic DNA isolation from *Escherichia coli* and designed primers for basic molecular biology applications.

Nichi-In Centre for Regenerative Medicine, Chennai, *Trainee - Laboratory Techniques* | May 2022

- Acquired practical exposure to Cell culture techniques, Class III Biosafety Cabinet operations and aseptic laboratory techniques.

ACHIEVEMENTS & FUNDING

- **UGC National Scholarship for Postgraduate Studies (NSPG), 2024–2026** -National merit-based scholarship awarded for postgraduate study.
- **Karunya Merit Scholarship, 2020–2024** - Merit-based scholarship awarded during undergraduate study.
- **IAESTE International Fellowship Award, 2025** - Competitively selected for an international experimental research internship at UCT Prague, Czech Republic.
- **Specialized Laboratory Techniques in Cell & Molecular Biology, IIT Madras, 2026** - Shortlisted candidate for advanced hands-on laboratory training.
- **Graduate Aptitude Test in Engineering (GATE) – Biotechnology, 2025** - Qualified.
- **NCRM NICHE QUIZ – Stem Cells & Regenerative Medicine** - Semifinalist in the national-level competition.
- **MSME Idea Hackathon 4.0** - Presented a biomedical innovation based on a smart hydrogel at the national level.

PRESENTATIONS

- **Invited Speaker**, IonCure Student Webinar & Lecture Series, IonCure Tech Pvt. Ltd, University of Colorado Colorado Springs & Healers Harbor (May 2026). Delivered a series of three lectures across interdisciplinary domains including Cancer Biology, Neuroscience, Drug Discovery, and AI-driven biological workflows.
- **Presenter**, International Conference on Advances in Molecular Medicine and Multiomics Technology (ICAMMT'25), PSG College of Technology, Coimbatore (Sept 2025) Presented research titled "*Inhalable Allicin Loaded Microparticle*," focusing on pulmonary drug delivery formulation and targeted therapeutic release.
- **Presenter**, EVOGEN'24 National Level Technical Symposium, Karunya Institute of Technology and Sciences, Coimbatore (Apr 2024)
Presented technical findings on "*Cyanocobalamin (Vitamin B12) Activity Quantification and Method Validation Using E. coli Mutant Microbiological Method in Micronutrient and Mineral Tablets*".
- **Participant**, Global Resilience Summit 2026, Healers Harbor & Lyda Hill Institute for Human Resilience (UCCS) (Apr 2026) Participated in interdisciplinary virtual sessions addressing AI, technological resilience, and healthcare innovation.
- **Participant**, 10th Annual Cell and Gene Therapy Symposium & Pre-Symposium Workshop, Centre for Stem Cell Research (BRIC-inStem / CMC Vellore) (Oct 2025). Attended specialized pre-symposium technical workshops and clinical translational sessions focused on advancements in cellular therapeutics and gene therapy pipelines.

REFERENCE

- Dr. Jissin Mathew (Master's Thesis Supervisor) - Karunya Institute of Technology and Sciences | jissin@karunya.edu
- Dr. Jibu Thomas (Head of the Department & Mentor)- Karunya Institute of Technology and Sciences | jibuthomas@karunya.edu
- Dr. Muthukumar Nadar (Course Instructor Molecular Biology & Immunology)- Karunya Institute of Technology and Science| muthukumar@karunya.edu